The Chinese University of Hong Kong, Shenzhen
School of Data Science

MAT3007 · Homework 4
Due: 11:59 pm, July 7, 2023

Instructions:

- Homework problems must be carefully and clearly answered to receive full credit. Complete sentences that establish a clear logical progression are highly recommended.
- You must submit your assignment in Blackboard. If some problems need coding, you should paste your code in the file. Please upload only **one** file(pdf). The file name should be in the format **last name-first name-hw4**.
- The homework must be written in English.
- Late submission will not be graded.
- Each student **must not copy** homework solutions from another student or from any other source.

Problem 1 (2 + 3 + 10 = 15pts). Cont. with Problem 5 in Homework 3.
Consider the following linear program:

$$\begin{array}{rc} \text{maximize} & 3x_1 + x_2 + 4x_3 \\ \text{subject to} & x_1 + 3x_2 + x_3 \leq 5 \\ & x_1 + 2x_2 + 2x_3 \leq 8 \\ & x_1, x_2, x_3 \geq 0 \end{array}$$

(a). What is the corresponding dual problem?

(b). Solve the dual problem graphically.

(c). Use complementarity conditions for the primal-dual pair to solve the primal problem.

Solution:

(a). The dual problem is

$$\begin{array}{rc} \text{minimize} & 5y_1 + 8y_2 \\ \text{s.t.} & y_1 + y_2 \geq 3 \\ & 3y_1 + 2y_2 \geq 1 \\ & y_1 + 2y_2 \geq 4 \\ & y_1, y_2 \geq 0 \end{array}$$

(b). The optimal solution is (2, 1) and the optimal value is 18.

(c). We first notice that the second constraint in the dual problem is not tight: $3y_1^* + 2y_2^* - 1 \neq 0$, which implies $x_2^* = 0$.

Then, we observe that the first and the third constraints are tight in the dual problem: $y_1^* + y_3^* - 3 = 0, y_1^* + 2y_2^* - 4 = 0$, therefore $y_1^* \neq 0, y_2^* \neq 0$, we have that

$$x_1^* + x_3^* = 5$$
$$x_1^* + 2x_3^* = 8$$

As a result, we get that $x_1^* = 2, x_3^* = 3$. Thus, the optimal solution of the primal problem is $(2, 0, 3)$.

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Problem 2 (5 + 10 + 10 + 10 = 35pts). Applications of LP adn Dual LP.
Consider the following table of food and corresponding nutritional values:

	Protein, g	Carbohydrates, g	Calories	Cost
Bread	4	7	130	3
Milk	6	10	120	4
Fish	20	0	150	8
Potato	1	30	70	2

The ideal intake for an adult is at least 30 grams of protein, 40 grams of carbohydrates, and 400 calories per day. The problem is to find the **least** costly way to achieve those amounts of nutrition by using the four types of food shown in the table.

(a). Formulate this problem as a linear optimization problem (specify the meaning of each decision variable and constraint).

(b). Solve it using MATLAB, find an optimal solution and the optimal value.

(c). Formulate the dual problem. Interpret the dual problem. (Hint: Suppose a pharmaceutical company produces each of the nutrients in pill form and sells them each for a certain price.)

(d). Use MATLAB to solve the dual problem. Find an optimal solution and the optimal value.

Solution:

(a). Let x_i denote the amount of food $i = 1, \cdots, 4$. The LP formulation is as follows:

$$\begin{aligned} \text{minimize} \quad & 3x_1 + 4x_2 + 8x_3 + 2x_4 \\ \text{s.t.} \quad & 4x_1 + 6x_2 + 20x_3 + x_4 \geq 30 \\ & 7x_1 + 10x_2 + 30x_4 \geq 40 \\ & 130x_1 + 120x_2 + 150x_3 + 70x_4 \geq 400 \\ & x_i \geq 0, \quad \forall i = 1, \ldots, 4 \end{aligned}$$

The first three constraints are the minimum daily value requirements for each nutrient.

(b). The optimal solution is (1.07, 0, 1.23, 1.08), and the objective value is 15.235 at optimal.

(c). Dual problem:

$$\begin{array}{ll} \text{maximize} & 30y_1 + 40y_2 + 400y_3 \\ \text{s.t.} & 4y_1 + 7y_2 + 130y_3 \leq 3 \\ & 6y_1 + 10y_2 + 120y_3 \leq 4 \\ & 20y_1 + 150y_3 \leq 8 \\ & y_1 + 30y_2 + 70y_3 \leq 2 \\ & y_i \geq 0, \quad \forall i = 1, \ldots, 3 \end{array}$$

Dual interpretation: A pharmaceutical company makes 3 kinds of pills. Each kind of pill contains 1 gram of a type of nutrient. The dual variable y_i represent the price for each pill of nutrient i. The company wants to sell the pills and maximize its profit. The objective function is the revenue of the company when consumers are buying the pills to satisfy their daily nutrient requirements. The constraints are restrictions on the prices of the pills to make it attractive to the customers. The total cost that can purchase nutrients equivalent to the food should not exceed the price of buying the food directly.

(d). The optimal solution is (0.310, 0.028, 0.012), and the objective value is 15.235 at optimal.

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Problem 3 (10 + 10 + 10 = 30pts). Max Flow Problem.
Consider the max flow problem on the graph in figure 1 with the orange node being the source node and the green node being the terminal node (the number on each edge is its capacity, see the lecture slide 13). Do the following based on the lecture slides.

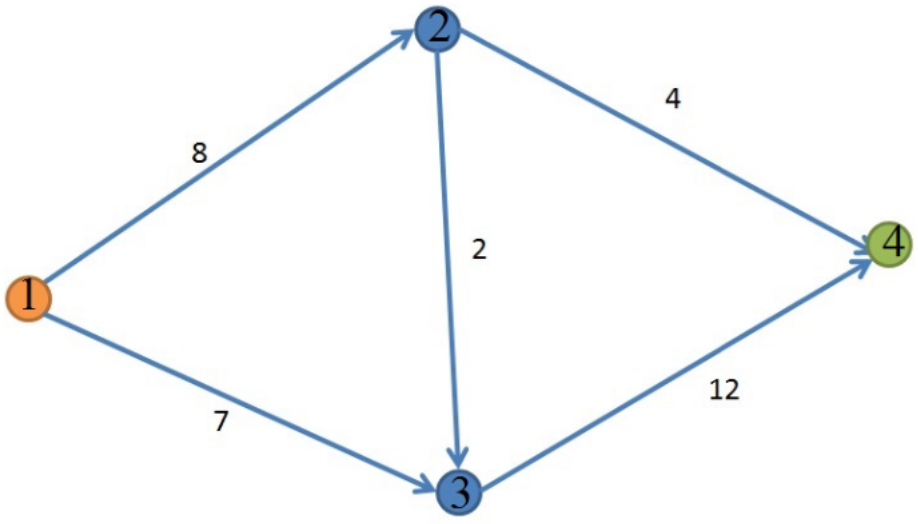


Figure 1: Max flow problem

(a). Formulate it as a linear program and solve it using MATLAB.

(b). Formulate the dual of this problem and solve it using MATLAB.

(c). Find the corresponding maximum flow and minimum cut of the graph.(Please draw the cut on figure 1).

Solution:

(a). Let x_{ij} denote the flow on edge (i, j). Let Δ denote the flow from imaginary node 0 to 1. The LP formulation of this problem is as follows:

$$\begin{array}{ll} \text{maximize} & \Delta \\ \text{s.t.} & x_{12} - x_{23} - x_{24} = 0 \\ & x_{13} + x_{23} - x_{34} = 0 \\ & \Delta - x_{12} - x_{13} = 0 \\ & x_{24} + x_{34} - \Delta = 0 \\ & 0 \le x_{12} \le 8 \\ & 0 \le x_{13} \le 7 \\ & 0 \le x_{23} \le 2 \\ & 0 \le x_{24} \le 4 \\ & 0 \le x_{34} \le 12 \end{array}$$

The optimal solutions are $x_{12} = 6, x_{13} = 7, x_{23} = 2, x_{24} = 4, x_{34} = 9$, and the objective value is 13 at optimal.

(b). Dual problem:

$$\begin{array}{ll} \text{minimize} & 8z_{12} + 7z_{13} + 2z_{23} + 4z_{24} + 12z_{34} \\ \text{s.t.} & z_{12} \ge y_1 - y_2 \\ & z_{13} \ge y_1 - y_3 \\ & z_{23} \ge y_2 - y_3 \\ & z_{24} \ge y_2 - y_4 \\ & z_{34} \ge y_3 - y_4 \\ & y_1 - y_4 = 1 \\ & z_{ij} \ge 0 \end{array}$$

Solving the problem using MATLAB, we get $y = (1, 1, 0, 0)$, $z = (0, 1, 1, 1, 0)$ and the optimal value is 13. The z_{ij} variables represent whether or not each edge is in the minimum cut.

(c). The cut is shown in figure 2.

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Problem 4 (10 + 10 = 20pts). Special Dual problem.
Suppose M is a square matrix such that $M = -M^T$, for example,

$$M = \begin{pmatrix} 0 & 1 & 2 \\ -1 & 0 & -4 \\ -2 & 4 & 0 \end{pmatrix}$$

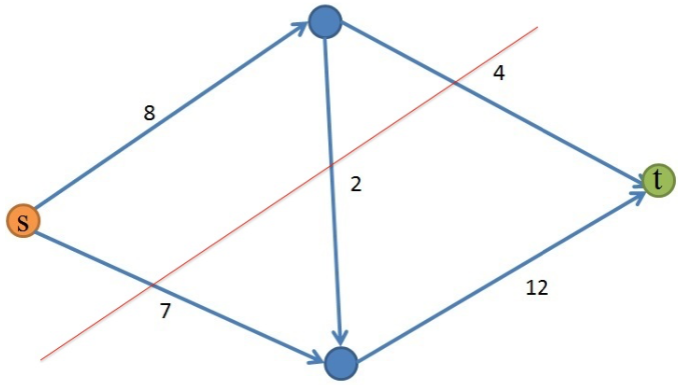


Figure 2: optimal cut

Consider the following optimization problem:

$$\begin{array}{ll} \text{minimize } \boldsymbol{x} & \boldsymbol{c}^T\boldsymbol{x} \\ \text{subject to} & M\boldsymbol{x} \geq -\boldsymbol{c} \\ & \boldsymbol{x} \geq 0 \end{array}$$

(a). Show that the dual problem of it is equivalent to the primal problem.

(b). Argue that the problem has optimal solution if and only if there is a feasible solution.

Solution:

(a). The dual is as follows:

$$\begin{array}{l} \max -\boldsymbol{c}^T\boldsymbol{y} \\ \text{s.t. } M^T\boldsymbol{y} \leq \boldsymbol{c} \\ \boldsymbol{y} \geq \boldsymbol{0} \end{array}$$

Since $M = -M^T$, $M^T y \leq c$ is equivalent to $My \geq -c$.
Therefore, we can reformulate the dual problem as a minimization problem as follows:

$$\begin{array}{l} \min \boldsymbol{c}^T\boldsymbol{y} \\ \text{s.t. } M\boldsymbol{y} \geq -\boldsymbol{c} \\ \boldsymbol{y} \geq \boldsymbol{0} \end{array}$$

Therefore, the dual problem is equivalent to the primal problem.

(b). First, it is obvious that if the problem has optimal solution, then it must have a feasible solution.
Now we prove the other direction.
If the problem has a feasible solution x, then $y = x$ is also feasible to the dual problem.
Therefore, both the primal and dual problems are feasible.
According to the weak duality theorem, they both have finite optimal solution.

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